

Lab 1 Activity Design

IT 213 Jan 16th 2026

Instructor: Bernal Manzanilla Saavedra

Title: Lab 1: Seeing Protocols in Action (Wireshark Intro) **Context:** Replaces or augments the "Intro to Packet Tracer" slot in Week 2. **Goal:** Satisfy syllabus outcome "Cover all layers in TCP/IP stack" by visually inspecting encapsulation.

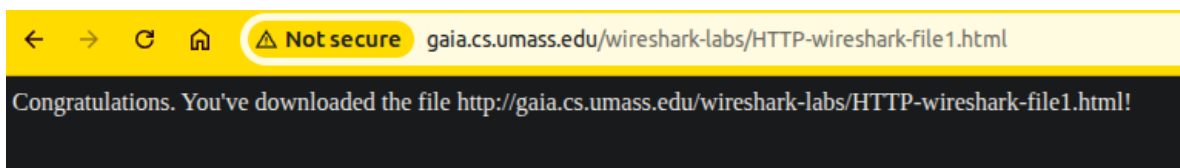
Instructions:

1. Preparation

- **Action:** Install Wireshark.
- **Context:** "As discussed in the syllabus, we need to understand the TCP/IP stack. Today we will 'sniff' the network to see these layers in reality."

2. The Capture (Mimicking Kurose)

- **Step A:** Open your web browser and clear your cache (or use "Incognito/Private" mode). **Close ALL other apps EXCEPT for the browser and wireshark**
- **Step B:** Open Wireshark and select your active internet interface (Wi-Fi or Ethernet).
- **Step C:** Start Capture (Click the Blue Shark Fin).
- **Step D:** In your browser, visit the following URL (specifically chosen because it is HTTP, not HTTPS, allowing us to see the text):
 - <http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html>



- *Note: Using a standard modern site (like Google) will result in encrypted QUIC/TLS traffic which is unreadable for this lab.*
- **Step E:** Once the page loads ("Congratulations!"), go back to Wireshark and click **Stop** (Red Square).

3. The Analysis (The "Aha!" Moment)

- **Filter:** In the top filter bar, type `http` and press Enter.
- **Find the Packet:** Look for a line that says `GET /wireshark-labs/INTRO-wireshark-file1.html`.

- **Inspect Encapsulation (The Stack):** Look at the "Packet Details" pane (bottom middle). They should expand the arrows to see the layers mapping exactly to your lecture topics:
 - **Frame:** Layer 1/2 (Physical/Link) – *Syllabus Topic: Link Layer*
 - **Ethernet II:** Layer 2 (MAC Addresses) – *Syllabus Topic: Link Layer*
 - **Internet Protocol Version 4 (IPv4):** Layer 3 (IP Addresses) – *Syllabus Topic: Network Layer*
 - **Transmission Control Protocol (TCP):** Layer 4 (Ports) – *Syllabus Topic: Transport Layer*
 - **Hypertext Transfer Protocol (HTTP):** Layer 5/7 (Application Data) – *Syllabus Topic: Application Layer*

4. Lab Questions (Deliverable)

Based on capture:

1. What is the **IP address** of your computer? (Found in the IP header Source field).
2. What is the **IP address** of the `gaia.cs.umass.edu` server? (Found in the IP header Destination field).
3. What **Transport Protocol** is being used? (TCP or UDP?)
4. **Syllabus Connection:** Take a screenshot of the "Packet Details" pane showing the 5 layers expanded. Label each layer with the corresponding name from our TCP/IP stack lecture.
5. Open a terminal and type “nslookup <IP that you found>